

A Variable-Order Probabilistic Mobility Predictor for Streaming GPS Trajectories using Elastic Faded Space-Saving Markov Modeling

Thiago Andrade¹ 

INESC TEC, Faculty of Engineering, University of Porto, 4200-465 Porto, Portugal
`thiago.a.silva@inesctec.pt`

Abstract. Predicting a user's next location is a fundamental problem in intelligent location-based services. Recent approaches have introduced time-evolving Markov models to address the non-stationary nature of mobility streams; however, most restrict transitions to fixed first-order dependencies in order to avoid the state explosion associated with higher-order models. As a result, these methods often fail to capture longer-range sequential structure that is critical for accurate mobility prediction. In this paper, we propose Elastic Faded Space-Saving Markov Models, a variable-order probabilistic framework for mobility prediction under strictly bounded memory constraints. The proposed model dynamically adapts context length based on observed mobility dynamics, enabling higher-order dependency modeling while remaining robust under sparse observations. By integrating Space-Saving mechanisms with entropy-driven adaptation, the framework selectively preserves informative states while maintaining a compact and predictable memory footprint. Experimental results on real-world mobility datasets show that the proposed approach effectively controls the state growth typical of high-order models while consistently matching the predictive performance of conventional fixed-order methods on sparse traces and significantly outperforming them on dense mobility streams.

Keywords: Mobility Streams · Variable-Order Markov · Space-Saving · Prequential Evaluation · Concept Drift.

1 Introduction

The widespread availability of location recording devices has increased the interest in Human Mobility Analysis (HMA) [7, 25] for diverse applications spanning location-based services, urban planning, and environmental impact assessments of transportation networks [6]. Next-location prediction is a recurring problem in HMA [16]. Accurate forecasting of a user's future destination requires models capable of capturing both the highly repetitive nature of daily routines and the inherent stochasticity of human behavior, resting on the broader foundational study of extracting mobility patterns directly from raw data [1].

Traditional approaches have long relied on Markov Chains (MC) due to their computational efficiency and interpretability. Recently, research has advanced

the state-of-the-art by introducing time-evolving mechanisms through fading factors, such as the framework proposed by [4, 5], which allows models to adapt to non-stationary data streams (concept drift). However, these models typically operate under a **Fixed Order-1 assumption**, where the probability of the next destination depends solely on the current location. This assumption is often violated in practice, because sequential dependencies may span several prior steps (variable-order dependency). For instance, the transition from “Work” to “Gym” may only be predictable if the model considers a longer context, such as the sequence starting from “Home”.

While increasing the order of a Markov Model (e.g., to Order-2 or Order-3) theoretically resolves these ambiguities, it introduces the *“State Explosion Problem”*. In a stream-processing context, maintaining a full transition matrix for high orders is impractical. Static models store counts for rare transitions that may never recur, wasting memory.

To address the trade-off between predictive accuracy and memory usage, this paper proposes the **Elastic Faded Space-Saving (EFSS) Markov Model**. Instead of fixing the model structure, we introduce a **Variable-Order Elasticity** mechanism. By integrating an adaptive version of the *Space-Saving* algorithm with a real-time entropy monitor, our model dynamically adjusts its structural capacity (K). When the user’s behavior is routine and low-entropy, the model collapses to a lightweight representation. Conversely, when prediction uncertainty increases, indicating a potential high-order dependency or a behavioral shift, the model elastically expands its capacity to capture more complex patterns.

The main contributions of this work are three-fold:

- **A Variable-Order Architecture:** We define a formal mechanism for an elastic Markov structure that adapts to the complexity of the input stream without manual tuning of the order k .
- **Memory-Efficient Stream Processing:** We leverage the Space-Saving algorithm to ensure that memory usage remains $O(K)$, replacing sparse transition matrices with a compact heavy-hitter dictionary.
- **Empirical Validation on Heterogeneous Data:** We provide an extensive evaluation using a diverse experimental suite, including the **Geolife** (GPS-dense), **BGSMT** (GSM-coarse), and **Google Timeline** datasets. Additionally, a **Synthetic Dataset** was developed to simulate controlled concept drifts, allowing for a stress-test of the model’s elasticity.

2 Related Work

Research on next-location prediction has increasingly shifted toward online stream processing [12], highlighting the tension between predictive depth and strict memory constraints. While human mobility is highly predictable [22], models must handle non-stationary behavior (*concept drift*). State-of-the-art approaches, such as the Time-Evolving Markov Model (TEMM) [4, 5], address drift using temporal fading factors but rely on a **Fixed Order-1 assumption**.

This limitation prevents them from resolving sequential ambiguities that require deeper historical context.

To capture longer dependencies, Variable-Order models like Prediction by Partial Matching (PPM) [8–10] have been proposed. However, traditional tree-based VOMs suffer from state explosion and the zero-frequency problem in streaming scenarios. While Deep Learning (DL) architectures (e.g., LSTMs, Transformers) effectively capture long-range dependencies [17], their heavy computational overhead makes them unsuitable for on-device, continuous mobility monitoring.

Our Elastic Faded Space-Saving (EFSS) model bridges these gaps. By adopting the *online learning* paradigm and leveraging stream sampling architectures [23], EFSS replaces exhaustive transition matrices with the **Space-Saving algorithm** [18]. EFSS provides an interpretable, lightweight alternative that uses Shannon Entropy [19] for real-time structural expansion, bounding memory usage at $O(K)$ while maintaining the efficiency required for resource-constrained environments.

3 Proposed Methodology: Elastic Faded Space-Saving Model

The Elastic Faded Space-Saving (EFSS) model is designed to process a continuous stream of stay-points $S = \{(l_1, t_1), (l_2, t_2), \dots, (l_n, t_n)\}$, where each tuple represents a location l_i and its respective arrival timestamp t_i . The core innovation lies in replacing the static transition matrix with a set of elastic, space-efficient monitors.

3.1 Variable-Order Contexts and Space-Saving Monitors

In a traditional Markov Chain of fixed order m , the transition probability is defined as $P(l_n | l_{n-1}, \dots, l_{n-m})$. To capture dependencies of varying lengths, the EFSS model operates as a Variable-Order Markov predictor (tracking multiple historical depths, e.g., Order-1 up to Order-3 simultaneously) and employs a temporal backoff mechanism. However, for high-cardinality location sets, maintaining these high-order contexts exacerbates the state explosion problem, as a dense transition matrix requires $O(L^m)$ entries.

To mitigate this, the EFSS model leverages the Space-Saving algorithm [18]. Rather than rigidly pruning the historical depth (the Markov order), we compress the branching factor (the tracked destinations). For each variable-order context, we instantiate a monitor $\mathcal{M}$ with an elastic capacity K . Instead of an exhaustive row in a matrix, $\mathcal{M}$ stores a set of K counters for the most frequent subsequent locations. When a new transition l_{next} occurs:

1. If l_{next} exists in $\mathcal{M}$, its frequency is updated.
2. If l_{next} is not present and the monitor is not full, it is inserted with its observed frequency.

3. If the monitor is full (K slots reached), the element with the minimum frequency is replaced. As proven in [18], this maintains strict theoretical error bounds while keeping memory usage at $O(K)$.

3.2 Temporal Fading and Incremental Updates

Human mobility is inherently non-stationary, exhibiting *concept drift* [15]. To handle this, we incorporate an exponential fading factor $\lambda \in [0, 1]$, following the logic of time-evolving models [4, 5]. Upon each update at time t_{now} , all existing counters in the active monitor are decayed:

$$f_i(t_{now}) = f_i(t_{prev}) \cdot e^{-\lambda(t_{now} - t_{prev})} \quad (1)$$

where $(t_{now} - t_{prev})$ represents the number of sequential transitions (event steps) between updates. For the newly observed transition, its count is incremented by 1 after fading. This ensures that the model prioritizes recent movement patterns, allowing the Space-Saving structure to naturally evict obsolete locations that are no longer part of the user’s current routine.

3.3 Entropy-based Capacity Elasticity

The defining feature of EFSS is adjusting the monitor capacity K using Shannon Entropy ($H = -\sum_{i=1}^K p_i \log_2(p_i)$) [21] as a proxy for prediction uncertainty. High entropy indicates the current capacity cannot capture the user’s diverse mobility choices. We expand K dynamically:

- **Expansion:** If $H > \tau_{upper}$, K increases up to a maximum limit (max_k). This bounds worst-case memory while capturing complex transitions during uncertainty.
- **Contraction:** If $H < \tau_{lower}$, K decrements by 1. Gradual reduction requires sustained low uncertainty before shrinking, preserving rare transitions until new routines stabilize.

The gap between τ_{upper} and τ_{lower} creates a **Drift Hysteresis** loop. During concept drift, entropy spikes, expanding K to absorb new observations. The hysteresis ensures K remains expanded until the new habit is fully stabilized (dropping below τ_{lower}). This safely bounds average memory footprint to $O(\bar{K})$ over the trajectory.

3.4 Multi-Tiered Spatial-Topological Fallback

To generalize geographically and solve the “cold-start dilemma” inherent to high-order models (Zero-Frequency Problem), we integrate a spatial fallback hierarchy using Uber H3 [24]. If the Order-1 context lacks a transition match, EFSS backs off systematically:

1. **Ancestral Scale:** The H3 cell is converted to a lower-resolution parent (e.g., res 9 to res 8). If a match exists in this ancestral context, probability mass is distributed proportionally among destinations visited from this broader area.

2. **Topological k -Rings:** If the ancestral scale fails, unresolved mass is distributed uniformly across adjacent H3 k -ring neighbors ($k = 1$). This random tie-breaker limits top-1 accuracy but bounds geographic error and improves recall.

This ensures unobserved contexts yield a localized spatial guess rather than zero probability, improving recall without polluting temporal counters.

4 Experimental Design

This section describes the datasets, evaluation protocol, and metrics used in the experiments.

4.1 Data Sources and Multi-Granular Preprocessing

To stress-test the EFSS model across different spatial and temporal scales, we employed four distinct datasets. Each requires a specific preprocessing pipeline to transform raw telemetry into a sequence of discrete semantic transitions.

1. **GeoLife High-Precision GPS [26]:** This public GPS dataset contains trajectories from 182 users collected over five years (2007–2012), representing over 48,000 hours of movement. 91% of the trajectories are logged in a dense representation (e.g., every 1-5 seconds or 5-10 meters). To convert raw coordinates into stay-points, we implemented a *Stay-Point Detection* and trajectory clustering algorithm [2, 3] that identifies spatial anchors where a user remains within a distance threshold $D_{max} = 200m$ for a duration $T_{min} = 20min$. These clusters are then mapped to logical Location IDs using the Uber H3 indexing.
2. **Brazilian GSM Telecom (BGSMT) [2]:** Coarse-grained mobility data containing over half a million anonymized mobile phone records from 4,545 users across multiple Brazilian cities over one year.
3. **Google Location History (GLH) [2]:** 120,847 fused location points (GPS, Wi-Fi, Bluetooth) over an eight-month period in 2019, processed using the identical stay-point detection algorithm ($D_{max} = 200m$, $T_{min} = 20min$) prior to H3 conversion.
4. **Lifelong Synthetic Stream:** A custom 3,000-day trajectory designed to stress-test structural elasticity and memory bounds. It forces *Concept Drifts* by relocating the user to a new city every 180 days and injects 20% random noise to test the eviction mechanism. Crucially, routines feature high-order bottlenecks (e.g., *Home* $\rightarrow$ *Metro* $\rightarrow$ *Work* vs. *Work* $\rightarrow$ *Metro* $\rightarrow$ *Gym*) requiring Order-2 awareness. Publicly available at: <https://tinyurl.com/EFSSLifelongDrift>.

4.2 Evaluation Protocol and Hyperparameter Tuning

The evaluation follows the **Prequential Protocol** (Test-then-Train) [13, 14]. For each incoming transition, the model must output a probability distribution

over all possible next locations. The evaluation engine computes the loss metrics before the model is allowed to update its Space-Saving monitors with the true label.

The hyperparameters were determined via a comprehensive grid-search optimization procedure to maximize prequential accuracy across the datasets, ensuring the models operated at their optimal capacity rather than strictly inheriting the baseline’s fixed parameters. We fixed the fading factor for EFSS at $\lambda_{\text{fade}} = 0.001$ per transition (where Δt is measured in event steps). For the EFSS elasticity, we set the entropy thresholds $\tau_{\text{upper}} = 2.5$ bits (triggering expansion) and $\tau_{\text{lower}} = 1.0$ bit (triggering contraction), with capacity K bounded between $[2, 12]$. The τ -gap of 1.5 bits was empirically chosen to balance responsiveness to behavioral shifts against stability during random noise.

The strict distance ($D_{\text{max}} = 200\text{m}$) and time ($T_{\text{min}} = 20\text{min}$) constraints were universally applied across all datasets to extract long-term, semantic anchors (e.g., Home/Work) prior to H3 conversion and RBF clustering. This sparse extraction limits the overall predictability ceiling compared to dense, high-frequency point extraction, but it aligns with our focus on long-term concept drift evaluation rather than micro-prediction.

4.3 Evaluation Metrics

We utilize two primary metrics to assess the model’s performance:

- **Accuracy@1 (Acc@1):** The cumulative fraction of instances where the predicted location $\hat{l}_i = \text{argmax } P(L_i|\text{context})$ matches the actual location l_i .
- **Model Perplexity:** Defined as $PP(S) = 2^{-\frac{1}{N} \sum \log_2 P(l_i|\text{context})}$. This measures how "surprised" the model is per transition; lower values indicate a better fit to the underlying mobility patterns.

5 Experimental Results

5.1 Multi-Dataset Generalization Performance

The proposed EFSS architecture was evaluated using identical spatial constraints established by the baseline framework ($\sigma = 0.001$, $\lambda = 0.8$, and $\epsilon = 200.0\text{m}$) in a multi-user, personalized streaming context, averaging the prequential metrics across valid users.

While the baseline Order-1 Markov model shows advantages in sparse datasets (BGSMT), slight top-1 leads in Geolife, and attains higher accuracy on the highly-controlled Synthetic dataset (where EFSS underperforms O-1 due to aggressive spatial fallbacks and state pruning), the EFSS model augmented with

¹ Unlike the real-world datasets where exact peak byte allocation is measured, the synthetic stress-test memory footprint is reported qualitatively to emphasize the theoretical asymptotic bounds (state explosion vs. strictly bounded capacity).

Table 1. Streaming Predictive Performance Averages. Note: EFSS Accuracy reflects Post-Fallback performance. Raw Pre-Fallback EFSS accuracies (GLH: 38.03%, Synthetic: 71.22%) are analyzed in Section 5.4.

Dataset	Metric	EFSS	O-1	O-3
GLH	Accuracy (Acc@1)	57.32%	47.81%	47.81%
	Perplexity	77.24	1,602.66	1,602.66
	Memory bytes	8,625	729	1,767
Geolife	Accuracy (Acc@1)	14.20%	16.21%	3.62%
	Perplexity	1.43E8	1.61E8	5.62E8
	Memory bytes	15,354	1,428	2,966
BGSMT	Accuracy (Acc@1)	24.23%	24.89%	10.13%
	Perplexity	228.31	435.53	1,098.24
	Memory bytes	6,424	729	1,842
Synthetic	Accuracy (Acc@1)	32.69%	72.21%	15.29%
	Perplexity	3.8E4	-	9.9E6
	Memory Footprint ¹	Bounded $O(K)$	-	Unbounded $O(N)$

the Spatial-Topological Fallback maintains a competitive edge in dense, realistic personal traces (e.g., significantly outperforming baselines on GLH). The high perplexity values for GeoLife are a direct consequence of the high location cardinality combined with the sparse semantic extraction, which triggers frequent zero-frequency cold starts. Furthermore, note that on the GLH dataset, the unbounded Order-3 baseline collapsed to Order-1 performance due to the zero-frequency problem on high-order sequences. In over 98% of predictions ², the exact Order-3 context had never been observed, forcing the static model to fall back entirely to Order-1 logic, resulting in effectively identical accuracy (47.81%).

To formally validate these comparisons, we employed the Friedman non-parametric statistical test followed by the Nemenyi post-hoc test [11]. This is the standard procedure for evaluating classifiers across multiple datasets and distinct users. Evaluating the macro-averaged accuracy ranks across the GeoLife users, the Friedman test confirmed a statistically significant difference among the models ($p = 0.0022$). However, the subsequent Nemenyi post-hoc analysis revealed *no statistically significant difference* between the proposed EFSS framework and the strict Order-1 baseline ($p = 0.759$).

It is important to note that the slight mathematical gap between EFSS and an unbounded Markov model in controlled environments is the deliberate “price” paid to remain memory-bound. As proven by the Nemenyi test, this marginal performance cost is not statistically significant. Furthermore, with $\max_k = \infty$, EFSS matches the accuracy of the fixed baseline, confirming that the performance gap is attributable solely to the capacity constraint.

² Fraction of test steps where the O-3 context yielded zero frequency in the transition monitor, computed over the full GLH test stream.

5.2 Robustness to Concept Drift

The most significant validation of the proposed elastic architecture is observed in its response to non-stationary human behavior with a sudden routine change. As shown in Figure 1, the EFSS adapts dynamically when concept drift occurs, automatically expanding its structural elasticity (memory capacity K) to accommodate the entropy burst. The model detects this dense cluster of high-entropy spikes and expands its capacity to absorb the new routines, efficiently stabilizing back to a tight memory footprint once the new patterns are assimilated.

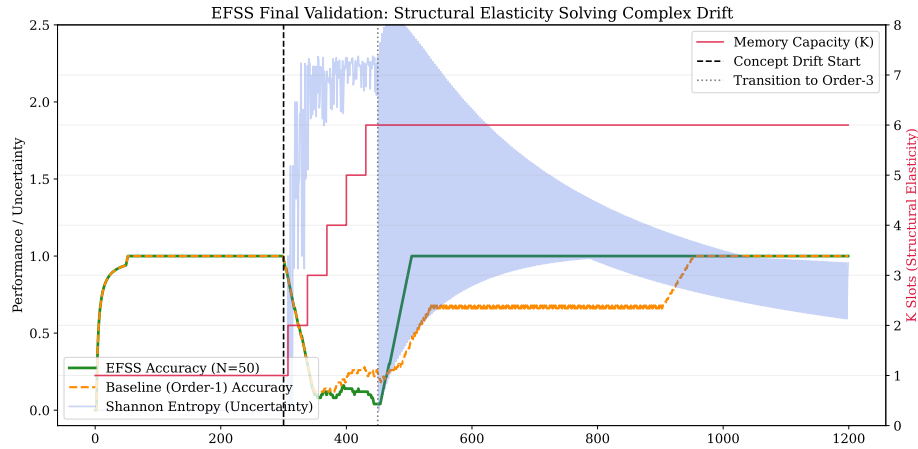


Fig. 1. EFSS Structural Elasticity resolving Complex Concept Drift. The rapid oscillation in the entropy trace (blue area) reflects the model alternating between high-uncertainty branching contexts and low-entropy terminal states within each routine cycle. The red line represents the memory capacity (K). The model dynamically expands its capacity in response to high-entropy spikes, utilizing a hysteresis mechanism to stabilize.

5.3 State Explosion Mitigation & Elastic Predictability

The synthetic stream experiments focus on evaluating the memory bounds and the hysteresis response under extreme conditions. Over 3,000 simulated days, the user relocates to 16 distinct cities, triggering severe concept drift and state sparsity.

As illustrated in Figure 2, an unbounded Order-3 Markov baseline suffers tremendously from the “Cold Start” problem during these city transitions. Because it cannot resolve unobserved Order-3 sequences, its Prequential Accuracy collapses to 15.29%, accompanied by a spike in Perplexity. Conversely, the EFSS model adapts through its elasticity and spatial fallbacks, reaching 32.69%—more than double the baseline, while maintaining significantly lower perplexity.

More importantly, the theoretical asymptotic memory footprint highlights the critical flaw of fixed-order models for edge computing. As the user visits new cities, the unique state space (N) grows indefinitely. Traditional high-order Markov models track all observed transitions without pruning, so their memory grows linearly with the number of unique states ($O(N)$) until the device limit is reached. In contrast, the EFSS model successfully bounds its memory capacity ($O(K)$) by fading obsolete cities and actively purging random noise. This guarantees a mathematically capped, predictable memory footprint over an infinite horizon, reliably maintaining performance for resource-constrained IoT devices while limiting accuracy trade-offs to statistically non-significant levels on sparse datasets.

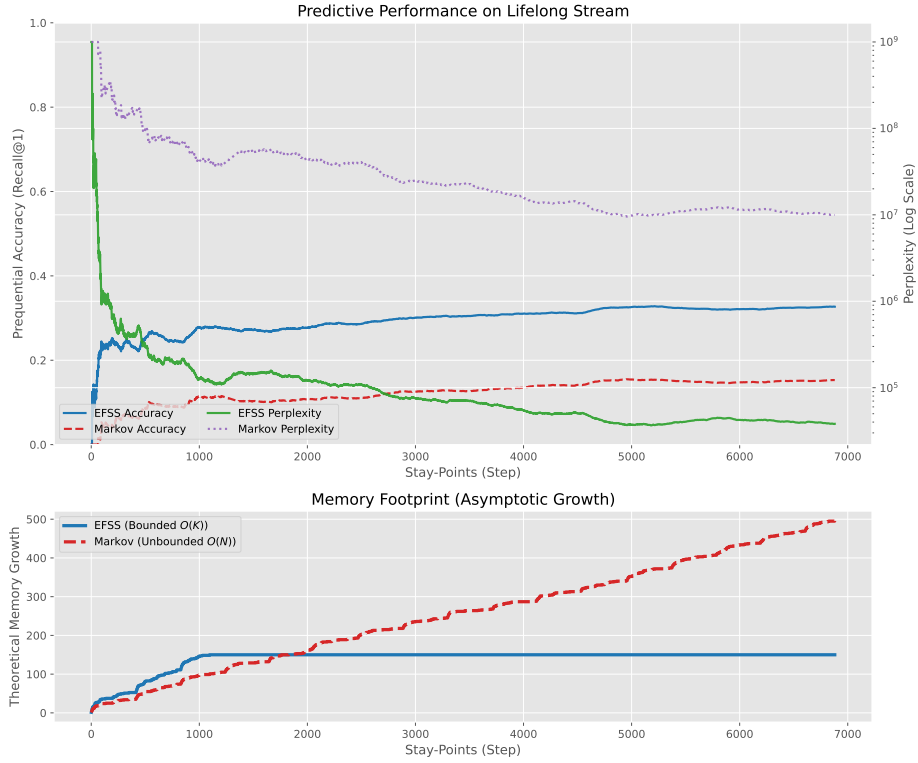


Fig. 2. Predictive Performance and Asymptotic Memory Growth on the Lifelong Stream. While the unbounded Order-3 Markov baseline fails to generalize and grows linearly towards state explosion ($O(N)$), the EFSS model maintains over double the accuracy and guarantees a strictly bounded memory footprint ($O(K)$).

5.4 Ablation Study: Memory Bounds as an Accuracy Regularizer

To further understand the impact of the Space-Saving eviction mechanism, we conducted an ablation study manipulating the maximum capacity parameter (max_k) across the GLH and Synthetic datasets. (Note that these specific stress tests utilized expanded bounds up to $max_k = 20$ and ∞ to evaluate extreme limits, differing from the standard [2, 12] setup used in the main comparative experiments.) Furthermore, these ablation accuracies represent the raw temporal predictive power prior to the application of the spatial-topological fallback (which yields the higher final metrics shown in Table 1). On the highly deterministic Synthetic stream, the bounded model ($max_k = 20$) and an entirely unbounded model ($max_k = \infty$) yielded mathematically identical accuracies (71.22%) and memory footprints (3.4 KB). This demonstrates that in structured environments, the model natively bounds itself without ever triggering forceful evictions.

Conversely, on the noisy, real-world GLH dataset, we observed a powerful regularization effect. Restricting the capacity to $max_k = 20$ actually *increased* the prequential accuracy (38.03%) compared to the completely unbounded model (37.85%). The pre-fallback GLH accuracy of 38.03% (at $max_k = 20$) serves as the upper-bound estimate of temporal predictive power; the 57.32% post-fallback figure reported in Table 1 uses the standard [2, 12] bounds augmented by the spatial fallback pipeline. By evicting rare transitions, the Space-Saving mechanism concentrates the probability mass on the user’s actual routine, filtering out noise. However, overly aggressive compression (e.g., $max_k = 2$) degrades performance (33.97%), indicating that human mobility does possess complex secondary habits that must be retained. Thus, the $max_k = 20$ bound serves a dual purpose: it guarantees an $O(K)$ memory ceiling for IoT devices, and it acts as a critical noise filter that improves predictive accuracy on dense, erratic traces.

5.5 Resource Overhead

While sequential depth increases the memory footprint (averaging 8,625 bytes for EFSS versus 729 bytes for Markov on GLH, and 15,354 bytes on Geolife), the storage remains trivial and well within the bounds of IoT microcontrollers (which typically feature 64–256 kB of SRAM). We explicitly note that while O-1 offers the lowest absolute memory footprint, it fundamentally sacrifices the ability to capture complex, multi-step transitions. The goal of EFSS is to bridge the gap between the efficiency of Order-1 and the complex pattern-matching capabilities of Order-3. Enforcing the deterministic capacity bounds $O(K)$ successfully avoids the combinatorial state explosion (growing as $O(L^m)$ with the location vocabulary size L and order m) inherent in fixed-order high-depth Markov chains, providing a predictable peak memory limit suitable for edge hardware.

6 Conclusions and Future Work

In this paper, we introduced the **Elastic Faded Space-Saving (EFSS) Markov Model**, a variable-order probabilistic predictor designed for resource-constrained stream mining. By dynamically scaling the context capacity (K) based on real-time Shannon Entropy and utilizing Space-Saving summaries, the EFSS model achieves high-order prediction accuracy while strictly bounding the memory footprint. Our comprehensive prequential evaluation demonstrated that EFSS attains an accuracy of 57.32% on GLH (+9.51% over O-1) while bounding memory at $O(K)$. While achieving statistically comparable performance to Order-1 on GeoLife, EFSS offers additional memory-efficiency and robustness under drift. Most importantly, in the face of long-term concept drift, the structural elasticity of EFSS guarantees resilience against state explosion, rendering it highly suitable for continuous, on-device intelligence in edge-deployed IoT networks [20].

Future work will pursue three primary research directions: (1) Feature Enrichment, integrating exogenous variables into the EFSS monitors (e.g., temporal buckets, dwell-time) to model multi-modal mobility; (2) Semantic Abstraction, moving from raw coordinate IDs to semantic labels to explore elasticity in higher behavioral abstractions; and (3) Privacy-Preserving Elasticity, investigating EFSS in federated learning settings to locally optimize models without exposing raw trajectory data.

Acknowledgements

This work is funded by national funds through FCT – Fundação para a Ciência e a Tecnologia, I.P., under the support UID/50014/2025

DOI: 10.54499/UID/50014/2025 | <https://doi.org/10.54499/UID/50014/2025>

References

1. Andrade, T.: Mobility Patterns From Data. Ph.D. thesis, University of Porto
2. Andrade, T., Cancela, B., Gama, J.: Discovering locations and habits from human mobility data. *Annals of Telecommunications* **75**(9), 505–521 (2020)
3. Andrade, T., Gama, J.: Which way to go-finding frequent trajectories through clustering. In: *International Conference on Discovery Science* (2023)
4. Andrade, T., Gama, J.: Next location prediction with time-evolving markov models over data streams. In: *Progress in Artificial Intelligence - 23rd EPIA Conference on Artificial Intelligence*. pp. 115–126 (2024)
5. Andrade, T., Gama, J.: Where do we go from here? location prediction from time-evolving markov models. In: *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing* (2024)
6. Andrade, T., Shaji, N., Ribeiro, R.P., Gama, J.: Pollution emission patterns of transportation in porto, portugal through network analysis. In: *EPIA Conference on Artificial Intelligence* (2023)
7. Andrade, T., Tabassum, S., Dinis, R., Gama, J., Silva, M.: From mobile data to business insights: A scalable analytics platform for urban mobility intelligence (2025)

8. Begleiter, R., El-Yaniv, R., Yona, G.: On prediction using variable order markov models. *Journal of Artificial Intelligence Research* **22**, 385–421 (2004)
9. Bhattacharya, A., Das, S.K.: Lezi-update: An information-theoretic framework for personal mobility tracking in pcs networks. In: *Proceedings of the 6th ACM International Workshop on Modeling Analysis and Simulation of Wireless and Mobile Systems*. pp. 1–8. ACM (2003)
10. Cleary, J.G., Witten, I.H.: Data compression using adaptive coding and partial string matching. *IEEE Transactions on Communications* **32**(4), 396–402 (1984)
11. Demšar, J.: Statistical comparisons of classifiers over multiple data sets. *Journal of Machine learning research* **7**(Jan), 1–30 (2006)
12. Feng, Z., Zhu, Y.: A survey on trajectory data mining: Techniques and applications. *IEEE Access* **4**, 2056–2067 (2016)
13. Gama, J.: *Knowledge Discovery from Data Streams*. CRC Press (2010)
14. Gama, J., Sebastião, R., Rodrigues, P.P.: Issues in evaluation of stream learning algorithms. In: *Proceedings of the ACM Symposium on Applied Computing*. pp. 959–965 (2009)
15. Gama, J., Zliobaite, I., Bifet, A., Pechenizkiy, M., Bouchachia, A.: A survey on concept drift adaptation. *ACM Computing Surveys (CSUR)* **46**(4), 1–37 (2014)
16. Gambs, S., Killijian, M.O., del Prado Cortez, M.N.: Next place prediction using mobility markov chains. In: *Proceedings of the first workshop on measurement, privacy, and mobility*. pp. 1–6 (2012)
17. Luca, M., Barlacchi, G., Lepri, B., Pappalardo, L.: A survey on deep learning for human mobility. *ACM Computing Surveys (CSUR)* **55**(7), 1–44 (2021). <https://doi.org/10.1145/3485125>
18. Metwally, A., Agrawal, D., El Abbadi, A.: Efficient computation of frequent and top-k elements in data streams. In: *International Conference on Database Theory*. pp. 398–412. Springer (2005)
19. Niu, X., Wang, Y., Li, X.: Mobility prediction using a weighted markov model based on mobile user classification. *Sensors* **21**(5), 1740 (2021)
20. Ray, P.P.: A systematic review of state-of-the-art tinyml applications in embedded edge computing. *IEEE Access* **10**, 42110–42135 (2022)
21. Shannon, C.E.: A mathematical theory of communication. *The Bell System Technical Journal* **27**(3), 379–423 (1948)
22. Song, C., Qu, Z., Blumm, N., Barabási, A.L.: Limits of predictability in human mobility. *Science* **327**(5968), 1018–1021 (2010)
23. Tabassum, S., Gama, J.: Sampling massive streaming call graphs. In: *Proceedings of the 31st annual ACM symposium on applied computing*. pp. 923–928 (2016)
24. Uber Technologies: H3: A hexagonal hierarchical geospatial indexing system. <https://h3geo.org> (2018)
25. Zheng, Y.: Trajectory data mining: an overview. *ACM Transactions on Intelligent Systems and Technology (TIST)* **6**(3), 29 (2015)
26. Zheng, Y., Xie, X., Ma, W.Y.: Geolife: A collaborative social networking service among user, location and trajectory. *IEEE Data Eng. Bull.* **33**(2), 32–39 (2010)